Artificial Intelligence (AI) Application To Art Therapy Using Drawings Of The HTP (House-Tree-Person) Test

by

Hajar Faiyad, 119200096
Ahmet Atakan Salar, 118200064

Supervised by

DR. Elena Battini Sönmez

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Abstract

House-Tree-Person (HTP) test is a projecting test intended to measure different aspects of personality where a person or a child is asked by the therapist to sketch a drawing of a house, a tree, and a person. This project aims to detect more markers for the House-Tree-person test by analyzing evaluated HTP test data and drawings provided by the Department Of Psychology of Istanbul Bilgi University using artificial intelligence, computer vision, and image processing techniques.

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List of Abbreviations

HTP	House-Tree-Person
CBCL	Child Behaviour Checklist
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
DL	Deep Learning
ML	Machine Learning
NLP	Natural Language Processing
AI	Artificial Intelligence
ResNet	Residual Neural Network

1 Introduction

Nowadays, it is a commonly agreed fact that the human brain is one of the most complex phenomena on earth. With passing generations, the interest in human psychology and how a human's mind works increases rapidly. Furthermore, with the influence of the current digital fast paced era we are living in, psychological awareness has gained popularity. Due to the commonly recurring crisis around the world, psychological illnesses and their symptoms have spread widely. Thus, psychological units and clinics became more crowded, which lead to the demand of more psychologists, psychiatrists, and therapists.

Although, in the field of psychology, there are numerous methods to identify a patient's mental condition, art therapy is considered one of the most widely used methods due to its efficiency to reflect a patient's unconscious brain activity. Usually, this method is used because it is hard to express one's unconscious thoughts through language or other communication methods. Moreover, unconscious thoughts seem to be expressed more evidently in drawings and arts. There are numerous methods to which art therapy could be conducted. Nevertheless, one tool of art therapy that is frequently used by therapists is the HTP test [1]. Standing for House, tree, and person, the test is mostly used in children who cannot express themselves due to lack if language or those who cannot express their feelings efficiently. From the test, therapists could evaluate and understand the child's psychological condition [2]. The HTP test seems to reflect potential psychological diseases, which are narrowed into an eight-syndrome checklist called the Child Behaviour Checklist, found in children aged 4 to 16 [3]. The child behaviour checklist (CBCL) entails diseases such as aggressiveness, withdrawal, and depression.

However, a therapist could only take a certain number of patients per day; therefore, to lighten the psychologist's tasks and to facilitate the diagnosis process, we would like to introduce an artificial intelligence algorithm that could extract new traits found in images related to the same behavioral cluster. Furthermore, the process of extracting the features from the drawings could be influenced by the therapist's personal psychological state. Although it is a fact that all humans are prone to error, an error in this form could lead to some unrecoverable consequences, such as using the wrong treatment method that would worsen a patient's condition. For these reasons, it is beneficial to maximize the number of psychological features markers to help psychologists remain neutral while proposing a possible diagnosis. In other words, Artificial Intelligence could eliminate any diagnostic biases coming from the psychologist's side.

The rest of the work is organized as follows. Section 2 overviews previous work done in this field. Section 3 outlines the design process. Section 4 summarizes the datasets used in the research. Section 5 generally introduces the methodology behind the project. In detail, subsection 5.1 overviews data preprocessing. Subsection 5.2 dives into image classification closely by outlining convolutional neural networks and transfer learning in subsections 5.2.1 and 5.2.2 respectively. Then, subsection 5.2.3 summarizes the image classification experiments done. Subsection 5.3 overviews psychological feature extraction from the HTP test, then subsection 5.3.1 summarizes the psychological feature detection experiments done so far. Finally, section 6 concludes the paper and introduces future work.

2 Related Works

The HTP test has already been used in [4] [5].

This section provides a quick overview of current studies on sketch images processing and the HTP test.

Kim et al. (2021) [4] first introduced the topic of image classification and symbolic feature extraction for the sketch images of the HTP test. They used a deep learning-based image captioning model for an art therapy support system. They proposed a research model consisting of three sub-processes: Object Classification, Psychological Feature Detection, and Caption Generation. Kim et al. (2021) used sketches of HTP sketches and a pre-trained model from a general image set to obtain a higher accuracy rate for object classification. Nevertheless, due to data limitations, the researchers did not utilize sketches of people. First, they labeled the objects as either 'tree' or 'house.' Then, they used the pre-trained model and ResNet algorithm, which can map inputs and outputs more efficiently by learning residual values and using skip connections to classify new images into predefined labels. They then obtained object-focused image vectors as output after freezing layers of the ResNet50 model to avoid sensitive changes and adding a CNN layer.

As for Psychological Feature Detection, Kim et al.(2021) first detected psychological objects such as windows and doors. They labeled each image with the frequency of each detected object. They input all of these images into a basic CNN model after assigning several features, or labels, to each drawing image so that the model can categorize all drawing images according to the characteristics. The output of this model anticipates being feature-focused picture vectors. Each image represents a matching vector that emphasizes any features that have already been discovered. Then they can get HTP picture vectors with information on psychological aspects by concatenating those two output vectors.

Lastly, for Caption Generation, Kim et al. (2021) employ an attention mechanism, a variation of the RNN-based seq2seq model, to identify significant elements in the entire drawing that include psychologically significant elements. Since the first two processes produce picture vectors with highlighted psychological traits, the attention mechanism may concentrate on the highlighted regions during caption creation. It can also identify significant sections of the input photos. However, to use attention mechanic=sim, they had to use the pre-trained image captioning model; Because the result of training represents the attention weight, which creates captions for HTP drawing pictures by filtering and choosing only highly weighted phrases from all parsed example captions of related terms for each image. Kim et al.(2021) finish their research by generating captions for HTP drawing images that may be used to diagnose psychological conditions.

Later on, Liu et al. (2021) [5] explored the research topic of image classification through the implementation of a convolutional neural network. In his research to train the model, 55 sets of control parameters are randomly chosen to create 55 distinct networks. The error between the model's actual test value and the predicted value given by the precision analysis formula is then compared. Through using a zero-mean Gaussian function, the initial weight value and deviation value may be randomly sampled. This random initialization has the purpose of dismantling symmetry. Even though their variation and numerosity, the convolutional neural network continually prompts numerous good algorithms. Liu et al. claimed that this approach is more practical since convolutional neural networks have made significant advances in picture features. As a result, the sector employs more convolutional neural networks. Using computers to interpret, analyze, and comprehend images allows for the technique known as "image classification," which assigns each image to one of several categories after identifying various targets and objects with varied patterns. The second is to employ a convolution neural network that could automatically discover picture attributes for image categorization. Convolutional neural networks directly utilize the information in an image's pixel information as input to develop hierarchical feature descriptions in either a supervised or unsupervised manner. An image's hierarchical properties may be combined using convolutional neural networks.

3 Design

In this section, we have the general steps that were followed in the project so far and the general plan for our future work. Figure 1 shows the project design overview and the overall plan.

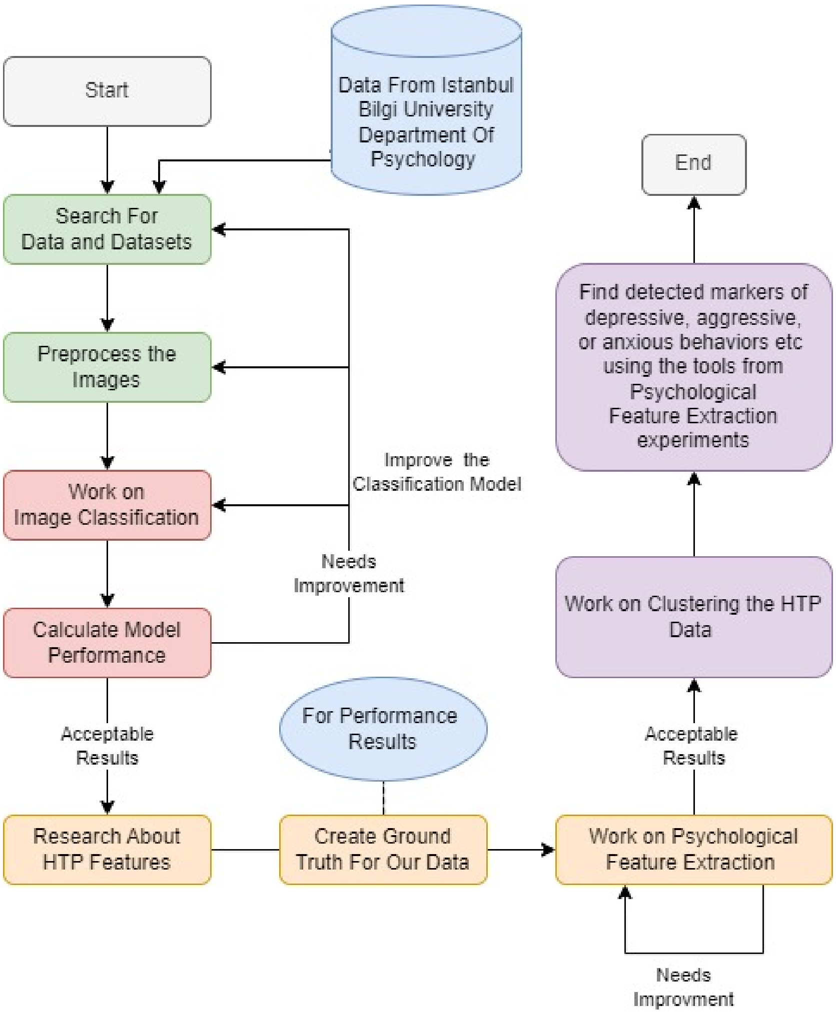

Figure 1: General project workflow plan overview

4 Datasets

There are two main datasets that we plan to use in the project. The crucial dataset among those is the images and data we received from the Department Of Psychology of Istanbul Bilgi University; these images are genuine House-Tree-Person test images drawn by children and analyzed by therapists, so they also contain analysis information with them.

However, considering the number of images in this dataset, we needed more image data, especially for Image Classification and some of the Psychological Feature Extraction tasks.

The Quick Draw Dataset [6], which is an image dataset shared publicly by Google, is a collection of 50 million drawings across 345 categories, contributed by players of the game Quick, Draw! Among the 345 categories of sketch images, we are only interested in house and tree categories. There are over 100,000 images each for these two categories in this dataset; however, there is no person class. This dataset is valuable because of the large number of sketch images it contains, but the similarity of these sketch drawings to our real HTP test data images is varied, as you can see by comparing the house sketch images in figure 2 and figure 4.

We might have enough house and tree class images for our image classification task, but we lacked person class images. To increase the number of person sketch images we have - we created some of our own person drawings only to be used in the image classification task.

Dataset Name	**House**	**Tree**	**Person**
Psychology Department HTP Test Data	61	61	135
Quick Draw! Dataset From Google	1000	1000	0
Our Own Hand-Drawn Person Images	0	0	216

Table 1: General statistics of all datasets used in the project

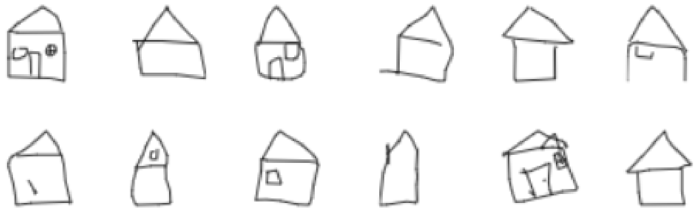

Figure 2: Example house drawings from Quick, Draw! Dataset from Google

5 Methodology

In this section, the methods that were used in the project will be discussed. Figure 3 shows the overall plan for the project output/result process pipeline.

Considering the nature of our problem and the overall aims and ambition of the project, there are multiple methods and algorithms used in the various steps. Our plan consists of using Deep Learning for Image Classification and Deep Learning and Image Processing for Psychological Feature Extraction. We will first talk about the data processing part, then the Image Classification Task, and later continue with our experiments in the Psychological Feature Extraction task so far.

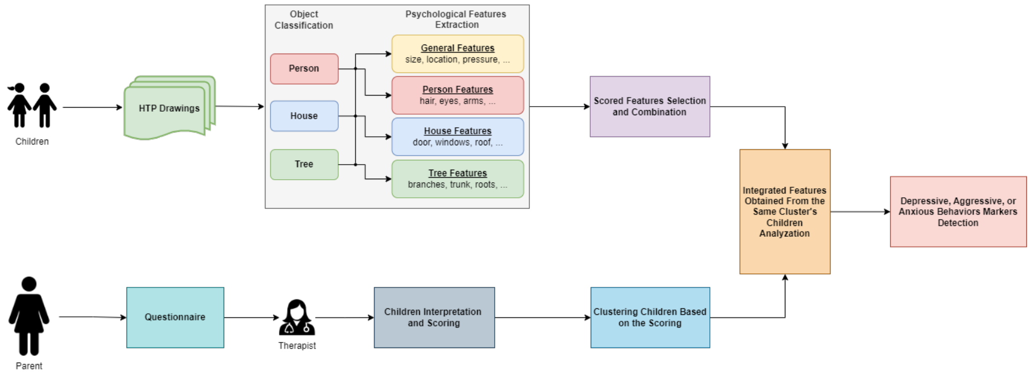

Figure 3: Overview of planned project output

5.1 Data Preprocessing

We received 257 real HTP Test images from the Department Of Psychology. This data contained the anonymous therapy analysis reports of the patients as well, such as their IQ scores and scores regarding their anxiety level (in the range of 0 to 18), depression (in the range of 0 to 26), and externalizing (aggressiveness) behaviour (in the range of 0-70). The image data we received was in PDF format, and they were manually scanned by the therapists with a smartphone camera. So preprocessing this image data is necessary.

The first step of preprocessing the data was to split each PDF file, which consisted of all drawings done by a child (house, tree, person), to separate files that only had a single drawing. We have received help from the working students in our University in this part.

After this process, we converted all one-image PDF files to PNG format images using a python script which could convert a PDF file to an image in PNG format.

During this process, we discarded a small number of images from our real HTP test data images because of the anomalies they contained. In the end, we had 54 house images, 55 tree images, and 56 person images to use as the test set in our image classification model experiments. The 79 person drawings from that dataset were added to our training set for the person class on top of our 216 hand-drawn person sketch images. So, we had 1000 house drawings, 1000 tree drawings, and 295 person drawings for our training set.

Figure 4: Example house, tree and person drawings from the real HTP Test Dataset

5.2 Image Classification

After preprocessing our real children's drawings images and collecting more drawings for each class, we had 2351 images; 1000 house drawings, 1000 tree drawings, and 351 person drawings

We decided to use Convolutional Neural Networks and transfer learning for our Image Classification task.

5.2.1 Convolutional Neural Networks (CNN)

Convolutional neural network (CNN) is one of the most popular and used of DL networks [7]. It is used in many different fields, such as image classification and object recognition and others. Its architecture consists of five connected structures, as shown in figure 5.

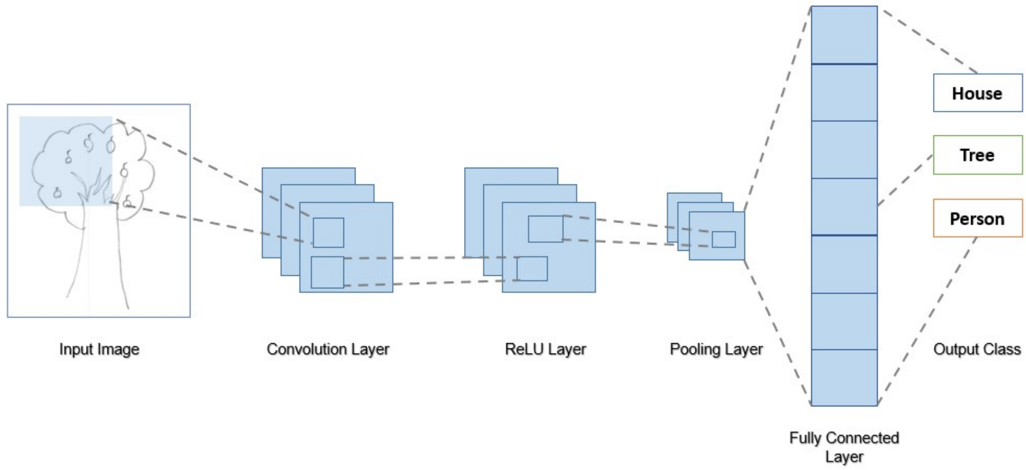

Figure 5: Example of a CNN Architecture for Image Classification

5.2.2 Transfer Learning

Transfer learning is a machine learning method where a model that was created for a particular task is later used again as the basis for a new model on a new problem. In deep learning fields like computer vision and natural language processing (NLP), fine-tuning a pre-trained model with the new data of another task is a popular method.

Fine-tuning is a very important part of the transfer learning approach because it allows our model to learn necessary new information required for the new task from the new data it is trained by. For example, in computer vision problems such as image classification or object recognition in images, there could be new classes or objects in the new task that did not exist in the initial data, so the model could not know the necessary information to classify or detect for that class or object. By fine-tuning the model, we make it possible for it to learn all the new information required for the new task.

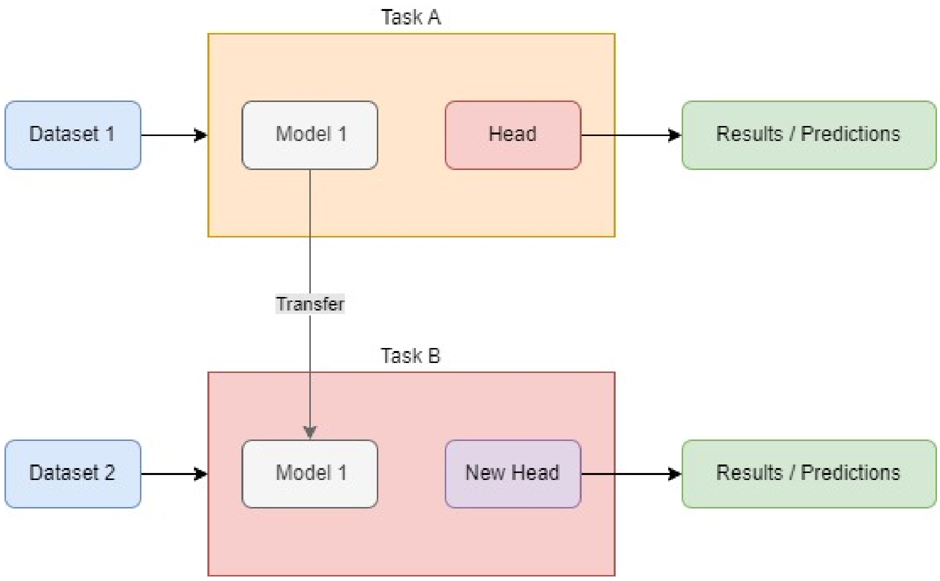

Figure 6: Transfer Learning Method Visualization

5.2.3 Image Classification Experiments

We decided to use the Resnet152 model that was introduced in the "Deep Residual Learning for Image Recognition paper" [8]. ResNet is a residual learning framework to ease the training of networks that are substantially deeper than those used previously.

By fine-tuning the ResNet152 model over our 3-class image dataset, we can get better results than we would if we trained our model from scratch. One of the main reasons for this decision was that we have an imbalanced dataset because of our lack of People class images.

There was also a considerable amount of difference between the images we had from the Quick, Draw! Dataset and our real HTP test dataset. This difference can be observed by comparing figures 1 and 4. To fix this problem, we had to apply pre-processing to our training and test images. We applied various pre-processing methods in our PyTorch implementation in fine-tuning, such as the random rotation method that randomly rotates the images and random perspective that randomly applies a perspective change effect to the images. We also increased the sharpness of the input images and resized all images to the same size.

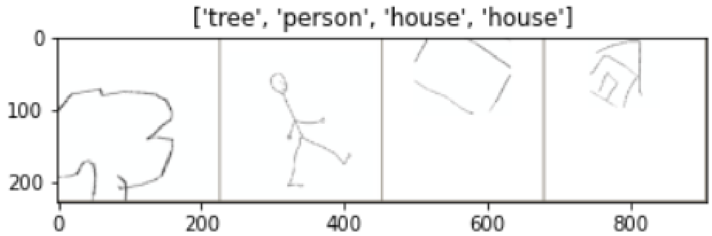

Figure 7: Example Train Set Images After Pre-Processing

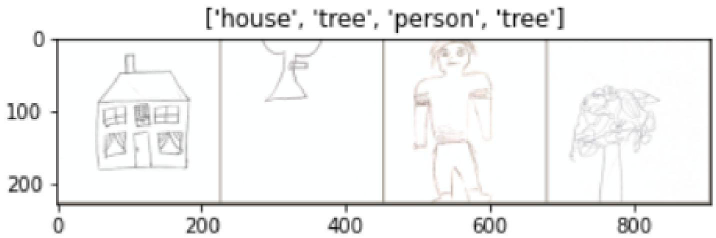

Figure 8: Example Test Set Images After Pre-Processing

After training the pre-trained Resnet152 model over our data in 10 epochs, we got a model with an accuracy of 66% over our House, Tree, and Person classes, so it could correctly predict only 110 images out of 165 test images. We are aware that the performance of our image classification model needs to be improved, but this is the first step of our experiments, and we will keep working on this task.

	Precision	**Recall**	**F1 Score**
0	0.83	0.65	0.73
1	0.54	0.91	0.68
2	0.81	0.45	0.57
accuracy	-	-	0.67
macro accuracy	0.73	0.67	0.66
weighted accuracy	0.73	0.67	0.66

Table 2: Performance report for our image classification experiment

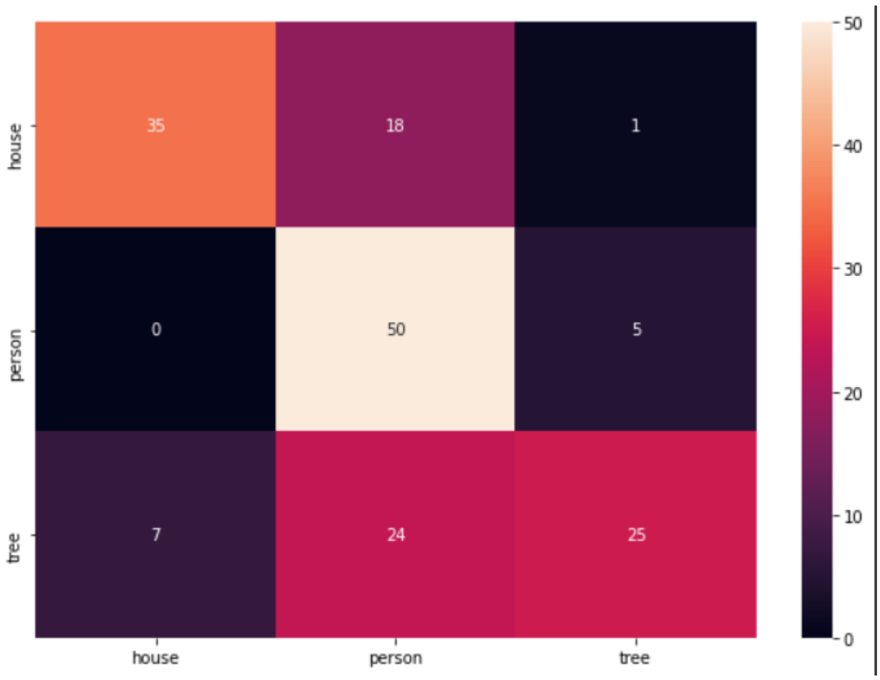

Figure 9: Confusion matrix for our image classification experiment

5.3 Psychological Feature Extraction

Feature extraction is the second task in our project. We have received a list of valuable features in the HTP test from the Department Of Psychology of Istanbul Bilgi University with the scoring table for these features. For example, one of the features that are analysed in the drawings is the Pen Pressure, which is how much pressure the child applied to the paper in the drawing, and there are five possible scores; 1 (very light), 2 (somewhat light), 3 (normal), 4 (somewhat heavy), 5 (very heavy).

Many features are valuable for the HTP test analysis by the therapists, and some of these features are more difficult to detect than others, so in this research, we will have to pick the most important features to work on. While picking this particular set of features, we asked for help from the Department Of Psychology. We have 27 features to work on possibly; 6 of them general (such as Pen Pressure, Drawing Size, etc.), 6 of them especially for house images (existence of roof, etc.), 12 of them for person class (such as the existence of eyes, nose, etc.), and 3 of them for tree class (such as the existence of trunks, etc.).

5.3.1 About Our Experiments

We have started to work on some of the easier features using image processing methods. For example, in the detection of the Pen Pressure feature, one of our experiments is to first apply pre-processing to the image, where we convert it to a black-and-white image. After this step, we calculate the number of black pixels in the drawing and then the white pixels for the background. After that, we apply the skeletonization method, which reduces all objects in the image to a skeletal remnant. After applying this method, we again calculate the number of black and white pixels in the image and find the black-pixel-to-white-pixel ratio of the image. In the last step, we compare this value with pre-defined ratio values to give it a value in the range of 1 to 5. If the ratio is large, it means the pen pressure is heavy.

We do not have any performance results for the feature extraction experiments; because we do not have the ground truth for these features yet. Without a ground truth for enough images, we cannot calculate the success of any of these experiments. We are working on creating a ground truth for our images with the help of working students from our university.

6 Conclusion And Future Work

House-Tree-person (HTP) test is a projecting test intended to measure different aspects of personality in children. In the evaluation of this test, there are some known markers, such as larger drawings correlating to aggressive behaviour in the child. We aim to detect more markers like this for the HTP test using AI and image processing techniques. To do this, we plan to cluster the HTP test data we received from the Department Of Psychology of Istanbul Bilgi University based on their similarities and extract common psychological markers from the drawing images.

Our tasks include creating an image classification model to detect the type of drawing, extracting psychological features from the images, and clustering & analyzing the processed images to detect new possible markers for the HTP test.

We have started our experiments in image classification, and we will keep working on more experiments to improve the performance of our model. We have initiated working on psychological feature extraction, and we are going to work on new features in the future. We are also working on creating a ground truth for these psychological features to test the performance of our experiments. When we have more concrete results on psychological feature extraction, we are going to use these tools to cluster and analyze the HTP test data to find common markers in the children behavioral clusters.

We have also found a new dataset [9] with 1000 person drawings drawn by children that we are planning to use in our new experiments to increase our performance in image classification and psychological feature extraction. This will be helpful as we had a lack of person drawings drawn by children in our datasets.

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